Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories to guide strategic business decisions on spending patterns, customer segments, and subscription behavior.

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Dataset at a Glance

3,900

Transactions

Rows in the dataset

18

Features

Columns of data

37

Missing Values

In Review Rating only

25

Product Types

Across 4 categories

Key features include customer demographics, purchase details, and shopping behavior (discounts, promo codes, frequency, ratings).

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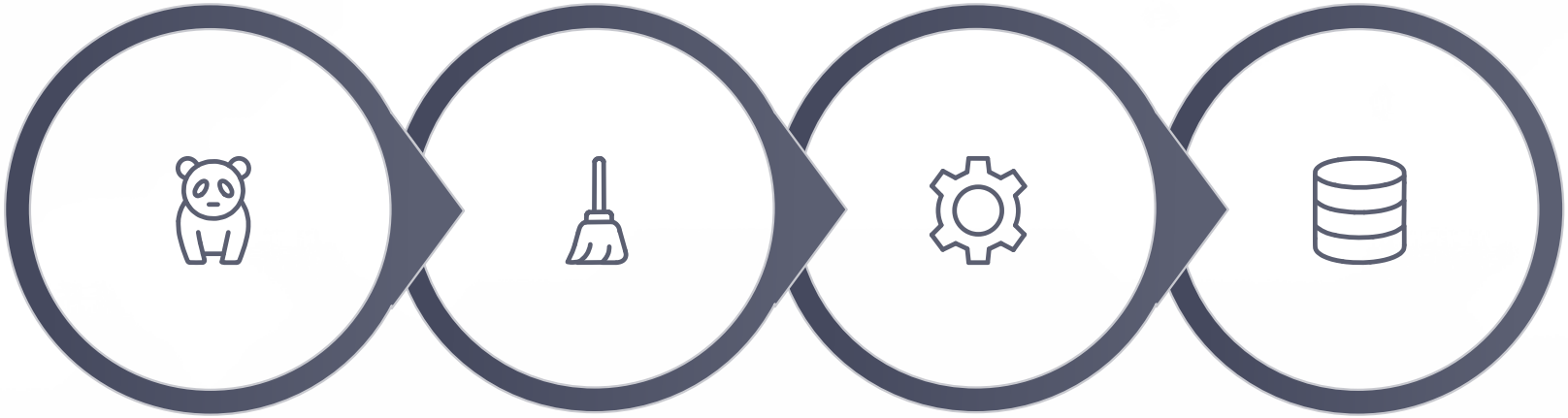
Data Preparation in Python



Load & Explore

Clean

Engineer Features

Load to PostgreSQL

Columns standardized to snake_case, promo_code_used dropped as redundant with discount_applied, and the cleaned DataFrame loaded into PostgreSQL for SQL analysis.

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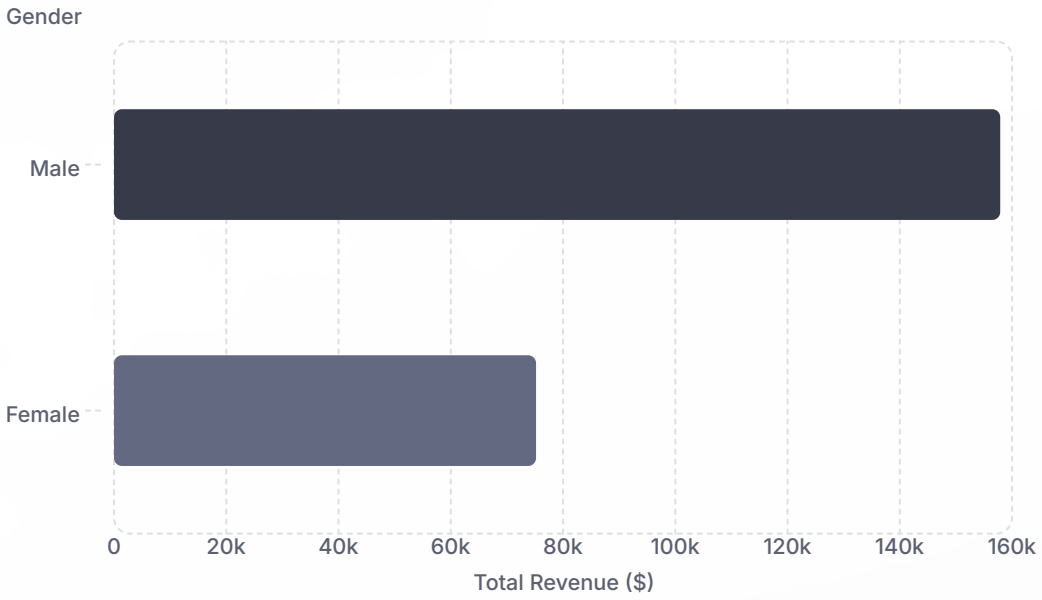
Revenue & Spending Insights



Key Findings

- **Male customers** generated 2× more revenue ($157,890 vs $75,191)
- **839 customers** used discounts and still spent above average
- Express shipping averaged **$60.48** vs Standard at **$58.46**
- Young Adults led revenue at **$62,143**, followed by Middle-aged ($59,197)

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Top-Rated Products by Category

Clothing

Blouse · Pants · Shirt

Top category by revenue with 1,737 purchases

Accessories

Jewelry · Sunglasses · Belt

Jewelry leads with 171 orders in this category

Footwear

Sandals · Shoes · Sneakers

Sandals top-rated in footwear with 160 orders

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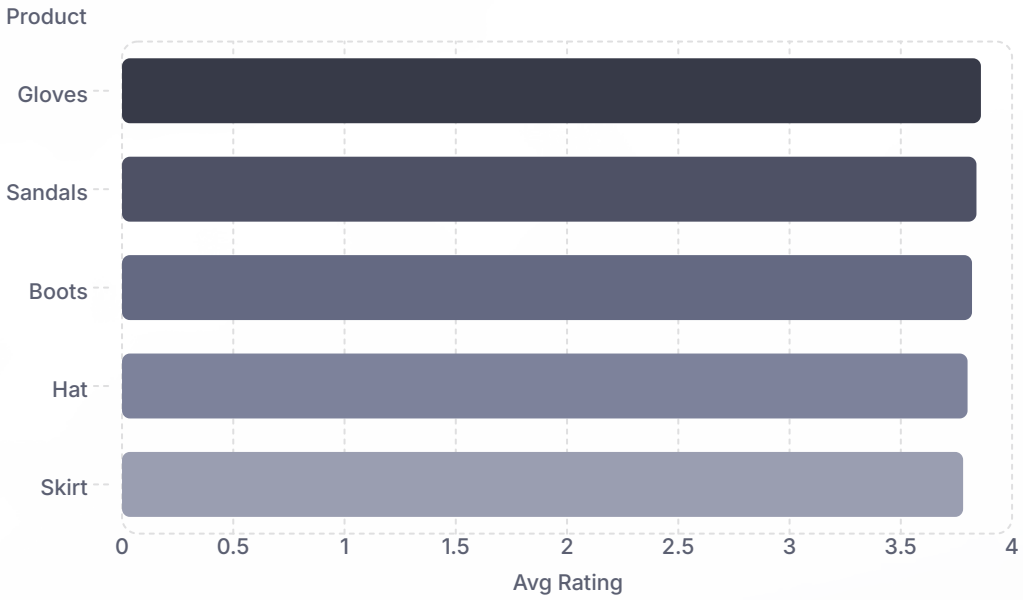
Product Ratings: Top 5



Overall Rating

Average review rating across all products: **3.75 / 5.0**

Footwear and accessories dominate the top-rated list, with Gloves and Sandals leading customer satisfaction.

Rating distribution: 25th percentile = 3.1, Median = 3.8, 75th percentile = 4.4

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Subscription Behavior

Non-Subscribers (73%)

2,847 customers · Avg spend **$59.87** · Total revenue **$170,436**

Subscribers (27%)

1,053 customers · Avg spend **$59.49** · Total revenue **$62,645**

Among customers with **5+ purchases**, 958 are subscribers vs 2,518 non-subscribers — a key opportunity to convert repeat buyers.

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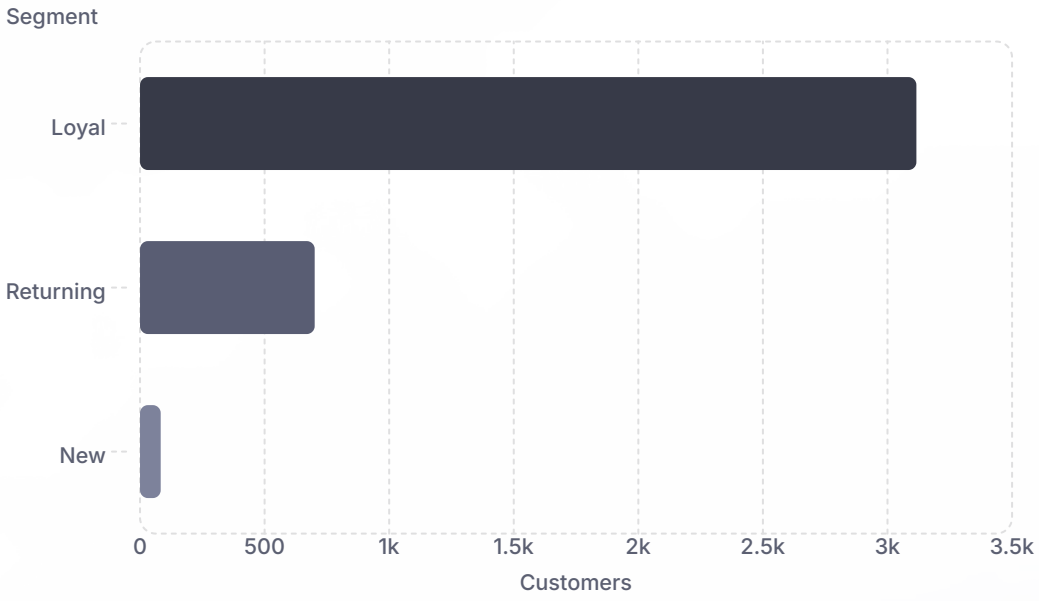
Customer Segmentation



Segment Breakdown

- **Loyal (80%)** — 3,116 customers with strong repeat purchase history
- **Returning (18%)** — 701 customers showing growing engagement
- **New (2%)** — 83 first-time buyers, high growth potential

Loyal customers dominate the base — focus on retention and upsell strategies.

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Power BI Dashboard

3.9K Customers

Total transaction records

$59.76 Avg Purchase

Mean spend per order

3.75 Avg Rating

Customer satisfaction score

Interactive filters for Subscription Status, Gender, Category, and Shipping Type enable drill-down analysis across revenue, sales, and age group visualizations.

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Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscriber base.

Loyalty Programs

Reward repeat buyers to grow the Loyal segment (currently 80% of customers).

Review Discount Policy

Balance sales boosts with margin control — 50% of Hat purchases used discounts.

Targeted Marketing

Focus on high-revenue age groups (Young Adults) and express-shipping users.

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